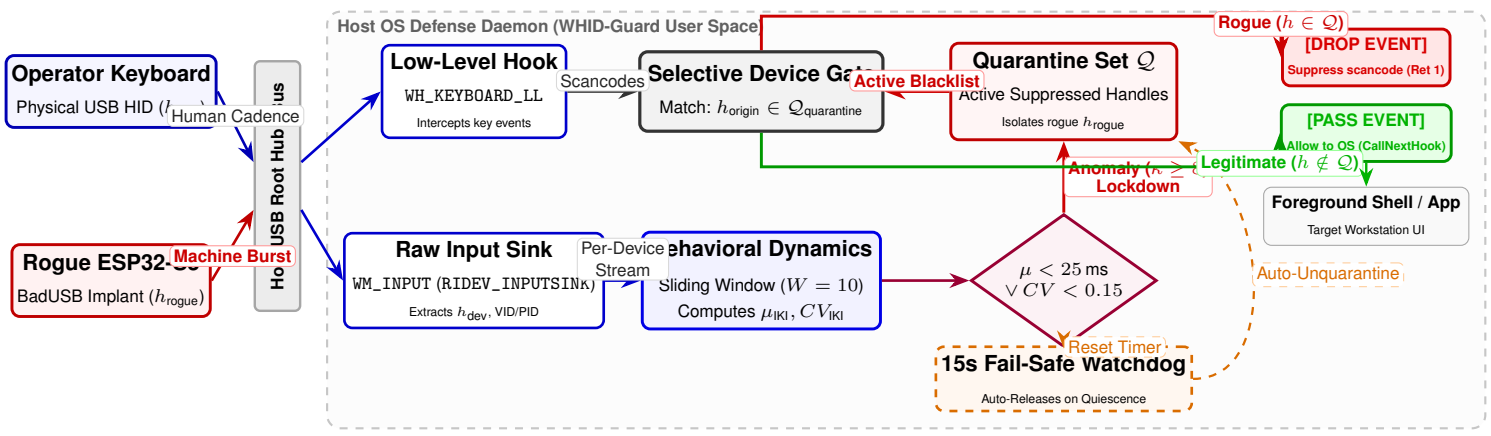

Host OS Defense Daemon (WHID-Guard User Space)
Operator Keyboard
Physical USB HID (h_{origin})
Human Cadence
Rogue ESP32-S3
BadUSB Implant (h_{rogue})
Machine Burst
Host USB Root Hub Bus
Low-Level Hook
WH_KEYBOARD_LL
Intercepts key events
Scancodes
Selective Device Gate
Match: $h_{\text{origin}} \in \mathcal{Q}_{\text{quarantine}}$
Active Blacklist
Quarantine Set $\mathcal{Q}$
Active Suppressed Handles
Isolates rogue h_{rogue}
Rogue ($h \in \mathcal{Q}$)
[DROP EVENT]
Suppress scancode (Ret 1)
[PASS EVENT]
Allow to OS (CallNextHook)
Legitimate ($h \notin \mathcal{Q}$)
Anomaly ($\kappa \geq$ Lockdown
Foreground Shell / App
Target Workstation UI
Raw Input Sink
WM_INPUT (RIDEV_INPUTSINK)
Extracts h_{dev}, VID/PID
Per-Device Stream
Behavioral Dynamics
Sliding Window ($W = 10$)
Computes $\mu_{\text{IKI}}, CV_{\text{IKI}}$
$\mu < 25$ ms
$\vee\, CV < 0.15$
Auto-Unquarantine
Reset Timer
15s Fail-Safe Watchdog
Auto-Releases on Quiescence